

Excel™ Add-In EPC Translator 1.7

B2		fx =FromSGTIN(\$A2,1)							
	A	B	C	D	E	F	G		
	RFID Tag	UPC/EAN	GTIN	Filter	Serial Number	Company Code	Length		
1					1213		7		
2	3014276DDC091E40000004BD	646007093371	00646007093371	0	14		7		
3	303627FDDC091E400000000E	9043831093378	09043831093378	1	67420		7		
4	3034676DDC091E800001075C	1694583093383	01694583093383	1	5981		7		
5	3074000000060E800000175D	00062022	00000000062022	3					
6									

BAIT Consulting
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Version Control

Ver	Date	Notes
0.9		Initial Version. SGTIN only.
1.0		SSCC, GRAI, GDTI, added additional info to SGTIN, added binary output.
1.1		Added GID functionality
1.2		Added BIN output
1.3		Misc fixes
1.4	7/15/2017	Added SGLN
1.5	9/10/2017	Fixed SGLN bug
1.6	5/3/2018	Added ToSGTIN2 – allows for using separate company code and item ref
1.7	8/30/2018	Fixed ToSGTIN2 – solved item ref error

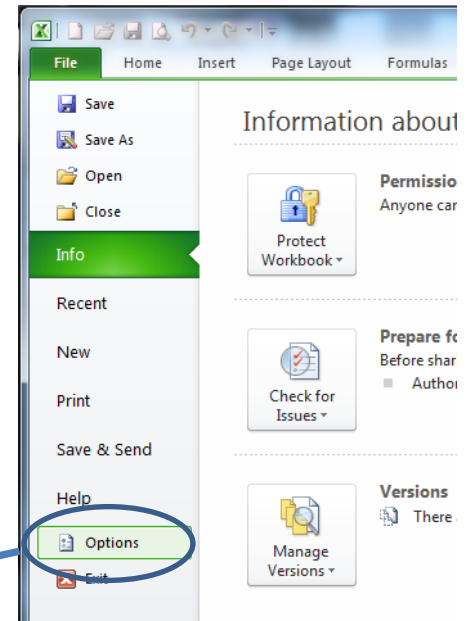
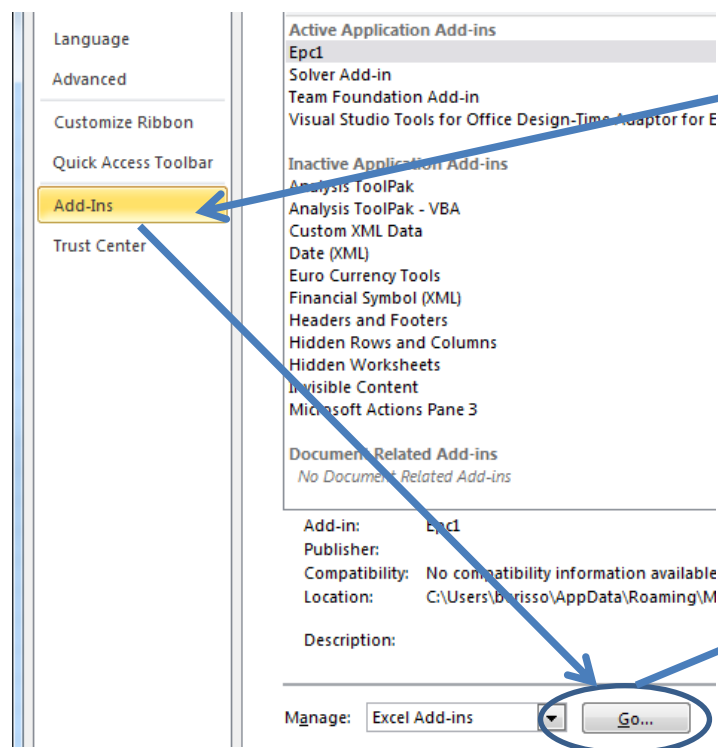
Introduction

With the increased usage of RFID in various industries, the need to be able to convert from the EPCglobal® family of standardized data encoding back to its base information (such as the originating GTIN or UPC/EAN value) has become more important. This Microsoft Excel Add-in has been designed to convert some of the common EPCglobal® formats back to their base parts. For a full discussion of how the conversions work, visit the GS1 EPCglobal® site (<http://www.gs1.org/epcglobal>).

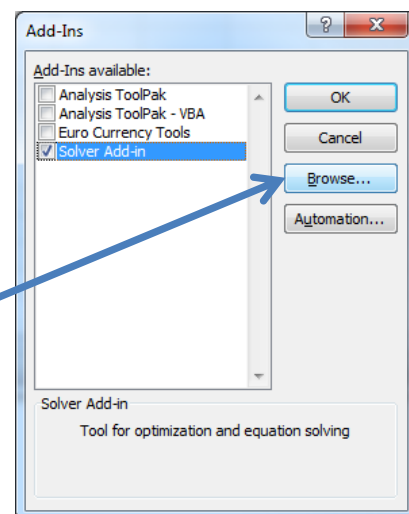
Installation

To install this add-in, open Microsoft Excel and click on the File tab (the round Office button for Excel 2007) and click on Options.

In the resulting dialog box, click on “Add-Ins” and then “Go”



When the “Add-Ins” dialog box appears, select “Browse” and select the “EPC_Translator_AddIn.xla” file.



The “EPC_Translator_AddIn” will be added to the list of available Add-ins.

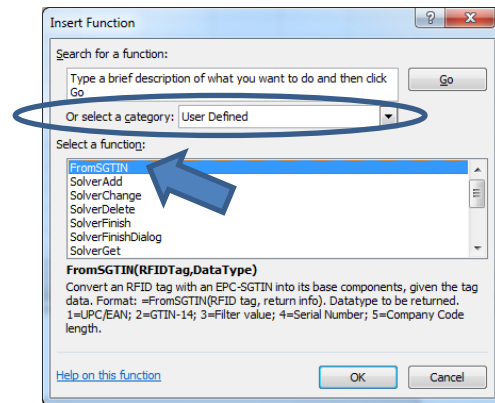
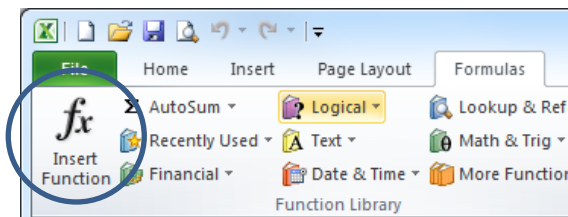
That’s it! You are now ready to translate your EPCglobal data into your desired format.

Usage

To use any of the EPC Translator functions, just type in the function name like you would with any other Excel function.

NOTE

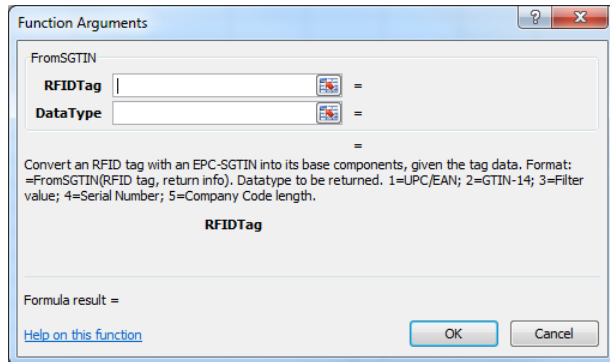
For users of Excel 2007, you will have to use the “Insert Function” button on the “Formulas” tab to access the EPC Translator functions. They will be found in the “User Defined” category.



FromSGTIN

To convert from a EPCglobal® Serialized Global Trade Item Number (SGTIN).

Pass the RFID tag's value as the first parameter and the information you want back as the second parameter.



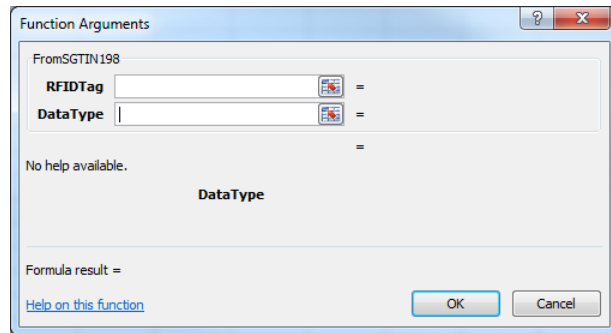
Data Type	Returned value
0	96 bit binary equivalent of the tag value.
1	UPC/EAN value (what is found on the product).
2	GTIN – the GS1 GTIN value that was actually encoded in the RFID tag.
3	Filter value (the unit of measure level per GS1 rules)
31	Filter value plus an abbreviated description
32	Filter value plus the full TDS description
4	Serial Number (the serial number of the item.
5	Company code length – the length of the company prefix per GS1 rules.
6	Company code (prefix) string
7	Item reference
99	Current date/time. Note that you have to format this cell for date and/or time or you will just receive a decimal number.

It should be noted that the function will return a string value for ALL datatypes. If you are using the returned value in a lookup function, you may need to convert the result from a string back to a numeric value. Use the “Value” function to do this.

FromSGTIN198

To convert from a 198 bit EPCglobal® Serialized Global Trade Item Number (SGTIN).

Pass the RFID tag's value as the first parameter and the information you want back as the second parameter.



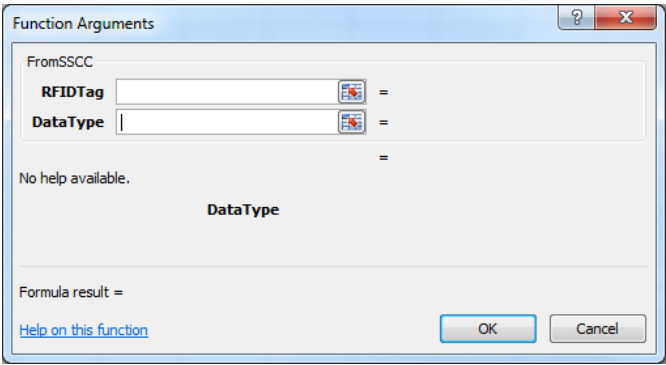
DataType	Returned value
0	198 bit binary equivalent of the tag value.
1	UPC/EAN value (what is found on the product).
2	GTIN – the GS1 GTIN value that was actually encoded in the RFID tag.
3	Filter value (the unit of measure level per GS1 rules)
31	Filter value plus an abbreviated description
32	Filter value plus the full TDS description
4	Serial Number (the serial number of the item.
5	Company code length – the length of the company prefix per GS1 rules.
6	Company code (prefix) string
7	Item reference
99	Current date/time. Note that you have to format this cell for date and/or time or you will just receive a decimal number.

It should be noted that the function will return a string value for ALL datatypes. If you are using the returned value in a lookup function, you may need to convert the result from a string back to a numeric value. Use the “Value” function to do this.

FromSSCC

To convert from a 96 bit EPCglobal® Serialized Shipping Container Code (SSCC).

Pass the RFID tag’s value as the first parameter and the information you want back as the second parameter.



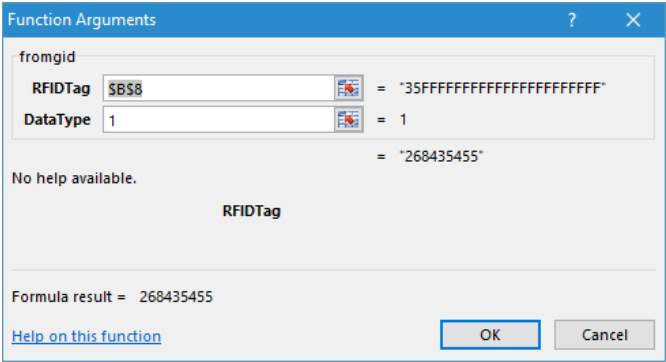
DataType	Returned value
0	96 bit binary equivalent of the tag value.
1	SSCC
3	Filter value (the unit of measure level per GS1 rules)
31	Filter value plus an abbreviated description
32	Filter value plus the full TDS description

It should be noted that the function will return a string value for ALL datatypes. If you are using the returned value in a lookup function, you may need to convert the result from a string back to a numeric value. Use the “Value” function to do this.

FromGID

To convert from a 96 bit EPCglobal® General Identifier.

Pass the RFID tag’s value as the first parameter and the information you want back as the second parameter.



DataType	Returned value
0	96 bit binary equivalent of the tag value.
1	General Manager Number
2	Object Class
3	Serial Number

It should be noted that the function will return a string value for ALL datatypes. If you are using the returned value in a lookup function, you may need to convert the result from a string back to a numeric value. Use the “Value” function to do this.

FromGRAI

To convert from a 96 bit EPCglobal® Global Returnable Asset Identifier (GRAI).

Pass the RFID tag’s value as the first parameter and the information you want back as the second parameter.

Function Arguments

FromGRAI

RFIDTag

DataType

|

No help available.

Formula result =

Help on this function

OK

Cancel

DataType	Returned value
0	96 bit binary equivalent of the tag value.
1	GRAI
2	Asset type
3	Filter value (the unit of measure level per GS1 rules)
31	Filter value plus an abbreviated description
32	Filter value plus the full TDS description
4	Serial Number
5	Company code length – the length of the company prefix per GS1 rules.

It should be noted that the function will return a string value for ALL datatypes. If you are using the returned value in a lookup function, you may need to convert the result from a string back to a numeric value. Use the “Value” function to do this.

FromSGLN

To convert from a 96 bit EPCGlobal Serialized Global Location Number (SGLN).

Pass the RFID tag’s value as the first parameter and the information you want back as the second parameter.

Function Arguments

FromSGLN

RFIDTag

↑

=

Datatype

↑

=

No help available.

RFIDTag

Formula result =

[Help on this function](#)

OK

Cancel

DataType	Returned value
0	96 bit binary equivalent of the tag value.
1	GS1 formatted GLN with extension.
2	GLN only
3	Filter value (the unit of measure level per GS1 rules)
31	Filter value plus an abbreviated description
32	Filter value plus the full TDS description
4	Serial Number
5	Company code length – the length of the company prefix per GS1 rules.
6	Company code (prefix) string
7	Location reference

It should be noted that the function will return a string value for ALL datatypes. If you are using the returned value in a lookup function, you may need to convert the result from a string back to a numeric value. Use the “Value” function to do this.

ToGID

To generate a GID tag value from the General Manager Number, Object Class and serial number.

Pass the component values as shown to get back hexadecimal tag value. As an optional parameter, you can set “CalcCheckDigit” to true and the GTIN check digit will be verified. If it is correct, “Check Digit OK” will be returned. If not, “Invalid Check Digit” will be returned.

Function Arguments

ToGID

GMN

Object

SN

ShowBool

=

No help available.

GMN

Formula result =

[Help on this function](#)

OK

Cancel

Data Type	Required Value
General Manager Number	Integer value between 0 and 2,68,435,455 inclusive.
Object Class	Integer value between 0 and 16,777,215 inclusive.
Serial Number	Integer value between 0 and 68,719,476,735.
Display Binary	Boolean option to display 96 bit binary equivalent of the tag value that defaults to False.

ToSGTIN

To generate a tag value from the GTIN, filter value and serial number.

Pass the component values as shown to get back hexadecimal tag value. As an optional parameter, you can set “CalcCheckDigit” to true and the GTIN check digit will be verified. If it is correct, “Check Digit OK” will be returned. If not, “Invalid Check Digit” will be returned.

Function Arguments

ToSGTIN

GTIN

=

Filter

=

CompanyCodeLength

=

SerialNumber

=

CalcCheckDigit

=

DisplayBinary

=

=

No help available.

CompanyCodeLength

Formula result =

[Help on this function](#)

OK

Cancel

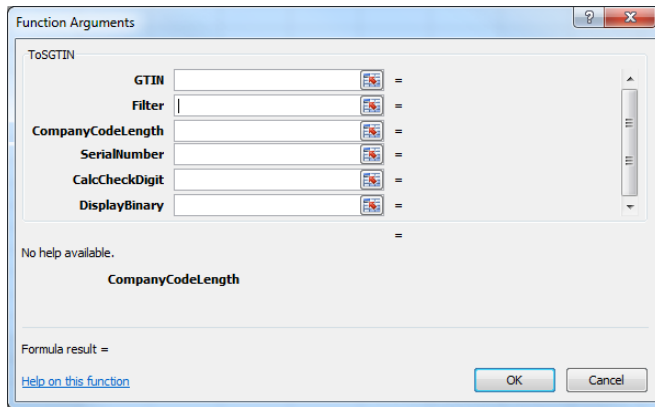
DataType	Required Value
GTIN	14 digit Global Trade Item Number as assigned or derived per GS1 rules.
Filter	Integer value of 0-7.
Company Code Length	Number of digits that make up the company code. Valid lengths are between 6-12
Serial Number	Integer value between 0 and 274,877,906,943
CalcCheck Digit	Boolean option to calculate the GTIN check digit that defaults to False.
Display Binary	Boolean option to display 96 bit binary equivalent of the tag value that defaults to False.

Note: CalcCheckDigit and DisplayBinary are not meant to work together. If both are set to True, CalcCheckDigit will take priority and the binary information will not be displayed.

ToSGTIN2

To generate a tag value from the GTIN, filter value and serial number.

Pass the component values as shown to get back hexadecimal tag value. As an optional parameter, you can set “CalcCheckDigit” to true and the GTIN check digit will be calculated. If it is correct, “Check Digit OK” will be returned. If not, “Invalid Check Digit” will be returned.



The image shows a 'Function Arguments' dialog box for the 'ToSGTIN' function. It contains several input fields: 'GTIN', 'Filter', 'CompanyCodeLength', 'SerialNumber', 'CalcCheckDigit', and 'DisplayBinary'. Each field has a small icon to its right. Below the fields, there is a section for 'No help available.' and a 'CompanyCodeLength' label. At the bottom, there is a 'Formula result =' field and a 'Help on this function' link. 'OK' and 'Cancel' buttons are at the bottom right.

DataType	Required Value
Company Code	GS1 assigned company code. Length is 6-12 digits.
Item Reference	Company assigned items reference number. Max length is 7 digits, but company code and item reference length combined cannot exceed 13 digits. Note – Item Reference MUST include indicator digit for GTIN as the first digit.
Filter	Integer value of 0-7.
Serial Number	Integer value between 0 and 274,877,906,943
CalcCheck Digit	Boolean option to calculate the GTIN check digit that defaults to False.
Display Binary	Boolean option to display 96 bit binary equivalent of the tag value that defaults to False.

Note: CalcCheckDigit and DisplayBinary are not meant to work together. If both are set to True, CalcCheckDigit will take priority and the binary information will not be displayed.

ToSGLN

To generate a tag value from the location, company code, filter value and extension.

Pass the component values as shown to get back hexadecimal tag value. As an optional parameter, you can set “CalcCheckDigit” to true and the GTIN check digit will be verified. If it is correct, “Check Digit OK” will be returned. If not, “Invalid Check Digit” will be returned.

[illegible]

Data Type	Required Value
Location	Location reference
Company Code	GS1 assigned company code
Company Code Length	Number of digits that make up the company code. Valid lengths are between 6-12
Filter	Integer value of 0-7.
Extension (optional)	Integer value between 0 and 2,199,023,255,551
Display Binary (optional)	Boolean option to display 96 bit binary equivalent of the tag value that defaults to False.

Note: If no extension is specified, a value of zero is implied. DisplayBinary requires that an extension of at least zero ("0") be added to the parameters.

ToSGLN2

To generate a tag value from the location, company code, filter value and extension.

Pass the component values as shown to get back hexadecimal tag value. As an optional parameter, you can set “CalcCheckDigit” to true and the GTIN check digit will be verified. If it is correct, “Check Digit OK” will be returned. If not, “Invalid Check Digit” will be returned.

Function Arguments

ToSGLN2

GLN

D12

↑

= "1234560012367"

Filter

D13

↑

= 0

CompanyCodeLength

D14

↑

= 6

Extension

↑

=

DisplayBinary

↑

=

= "32180000000000000000000000000000"

No help available.

GLN

Formula result = 32180000000000000000000000000000

Help on this function

OK

Cancel

DataType	Required Value
GLN	Valid GS1 GLN
Filter	Integer value of 0-7
Company Code Length	Number of digits that make up the company code. Valid lengths are between 6-12
Extension (optional)	Integer value between 0 and 2,199,023,255,551
Display Binary (optional)	Boolean option to display 96 bit binary equivalent of the tag value that defaults to False.

Note: If no extension is specified, a value of zero is implied. DisplayBinary requires that an extension of at least zero (“0”) be added to the parameters.

Trouble Shooting

Problem	Solution(s)
Function returns “Evaluation version has expired”	Your evaluation version of the Add-in has expired. Please contact BAIT Consulting (www.bait-consulting.com) or your distributor for a licensed version of the add-in.
Function returns “Invalid company code length”	The encoded data in your RFID tag is not SGTIN compliant. Specifically, the calculated company code does is not the correct length.
Your VLookUp is not returning a valid response.	The function is returning a string value for the value of what you are looking up. If your lookup value should be formatted as a number, use the “Value()” function in Excel to convert the text back to a numeric value.
You get a message box stating an “Invalid hex character found:”	There are characters in the RFID tag string that are not valid hexadecimal values. Check your data string.
Function returns “#Value!”	Your RFID tag string is not valid. Check your data string.
You have spaces in your tag data	Use Excel’s “Substitute” function to remove the spaces from in between sets of digits. =Substitute(A1,” “,””)

